PricebookCalc Developer Guide

This guide documents every feature in PricebookCalc, explaining how each one works, where the code lives, and how the pieces connect.

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1. Architecture Overview

Tech Stack

Layer	Technology
Frontend	React 18, TypeScript, Vite
Routing	Wouter (client-side)
State	TanStack React Query (server), React Context (cross-calculator)
UI	shadcn/ui (Radix primitives), Tailwind CSS
Charts	Recharts
Backend	Node.js, Express 5, TypeScript (ESM)
Database	PostgreSQL (Neon-backed via Replit)
ORM	Drizzle ORM with drizzle-zod validation

Project Structure

```
client/src/
   App.tsx                    # Root component, routing, providers
   contexts/
      CalculatorContext.tsx   # Shared calculator state (System + Repair)
      ServiceTitanAuthContext.tsx  # ServiceTitan password auth state
      AdminAuthContext.tsx    # Admin PIN auth state
```

```
  components/
     Navigation.tsx          # Sidebar navigation with admin panel
     ResultsCard.tsx         # Shared results display (price, GM%, chart)
     PasswordGate.tsx        # Password prompt wrapper for protected pages
     VersionCheck.tsx        # New-version-available banner
     InputMoney.tsx          # Reusable currency input component
  pages/
     Calculator.tsx          # System Calculator page
     RepairCalculator.tsx    # Repair Calculator page
     ReverseCalculator.tsx   # Reverse Calculator page (System & Repair modes)
     PushToServiceTitan.tsx # Push page with 3 tabs
     PushHistory.tsx         # Push history log
     Tenants.tsx             # Tenant management (CRUD)
     Locations.tsx           # Location/sales-tax management (CRUD)
     Settings.tsx            # Global calculator settings
     History.tsx             # Saved quotes list
  hooks/
     use-settings.ts         # Settings fetch hook with localStorage cache
  lib/
      queryClient.ts          # TanStack Query client setup

server/
  index.ts                   # Server entry point
  routes.ts                  # All API route handlers (~1,300 lines)
  storage.ts                 # Database CRUD operations (Drizzle)
  db.ts                      # Database connection
  servicetitan.ts            # ServiceTitan API client class
  pushProcessor.ts           # Background push processing logic
  pushQueue.ts               # Push queue with concurrency control
  pushProgress.ts            # Server-Sent Events for push progress
  vite.ts                    # Vite dev server integration
  static.ts                  # Static file serving (production)

shared/
  schema.ts                  # Drizzle table definitions, Zod schemas, types
  routes.ts                  # Shared API route definitions with Zod
```

Key Design Patterns

- **Shared Route Definitions**: API routes defined once in `shared/routes.ts` with Zod schemas — used by both client and server for type-safe API calls.
- **Settings Singleton**: The `settings` table has exactly one row. Auto-created with defaults if missing.
- **Quote Snapshots**: Saved quotes capture calculated results at the time of saving, preserving history even if global settings change later.
- **CalculatorContext**: React Context that holds both System Calculator state and Repair Calculator state, enabling cross-calculator data sharing (e.g., "Copy from" in Reverse Calculator).

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2. Database Schema

File: `shared/schema.ts`

Tables

`settings` (Singleton) Stores all global calculator configuration. One row (id=1), auto-created with defaults.

Column	Type	Default	Purpose
`id`	serial	auto	Primary key
`targetMePercent`	numeric	"23.00"	Default M&E target percentage
`salesTaxPercent`	numeric	"6.35"	Default sales tax rate
`laborMode`	text (enum: "hours"/"percent")	"percent"	Which labor input mode to use
`laborHourlyRate`	numeric	"0.00"	Hourly rate when labor mode is "hours"
`laborPercent`	numeric	"7.00"	Labor % when mode is "percent"
`laborBurdenPercent`	numeric	"20.00"	Employer burden applied to labor and commission
`commissionPercent`	numeric	"10.00"	Commission as % of selling price
`commissionBurdenPercent`	numeric	"20.00"	Burden applied to commission
`financingFeePercent`	numeric	"6.00"	Financing fee as % of price
`grossMarginMinPercent`	numeric	"52.00"	Minimum GM% threshold for health indicators
`serviceTitanPassword`	text	"ZephyrHome160"	Password for ServiceTitan section access
`adminPin`	text	"7437"	PIN for Admin section access

`quotes` Saved calculation snapshots from the System, Repair, and Repair GM calculators.

Column	Type	Default	Purpose
`id`	serial	auto	Primary key
`title`	text	"Untitled Quote"	User-given quote name
`materialsCost`	numeric	required	Input: materials cost
`equipmentCost`	numeric	required	Input: equipment cost
`laborHoursInput`	numeric	"10"	Input: labor hours (hours mode)
`calculatedPrice`	numeric	required	Snapshot: final calculated price
`calculatedGrossMargin`	numeric	required	Snapshot: GM% at save time
`calculatedEstimatedLabor`	numeric	required	Snapshot: estimated labor at save time
`calculatedMePercent`	numeric	required	Snapshot: M&E% at save time
`subcontractorCost`	numeric	"0.00"	Input: subcontractor cost
`craneCost`	numeric	"0.00"	Input: crane cost
`permitCost`	numeric	"0.00"	Input: permit cost
`userId`	integer	null	FK to `users` — who saved the quote

Column	Type	Default	Purpose
customerName	text	null	Optional customer name (captured via Reveal Price gate)
tenantId	integer	null	FK to tenants, ON DELETE SET NULL — which tenant/business unit this quote was calculated for, based on the tenant selected in TenantTaxSelector at save time. Nullable — quotes saved before this column existed have tenantId = null and cannot be backfilled.
createdAt	timestamp	now()	When quote was saved

Note: tenantId is used purely for client-side filtering on the Saved Quotes page (see Section 10) — no backend query filters on it directly; role-based tenant scoping happens via GET /api/tenants.

tenants ServiceTitan API credentials for multi-tenant support.

Column	Type	Default	Purpose
id	serial	auto	Primary key
name	text	required	Display name
tenantId	text	required	ServiceTitan tenant ID
clientId	text	required	OAuth client ID
clientSecret	text	required	OAuth client secret
appKey	text	required	ServiceTitan app key
locationId	integer	null	FK to locations table (for sales tax lookup)
isActive	boolean	true	Whether tenant is enabled
sortOrder	integer	0	Display ordering
createdAt	timestamp	now()	Creation timestamp
updatedAt	timestamp	now()	Last update timestamp

locations Reference library mapping location names to sales tax percentages.

Column	Type	Default	Purpose
id	serial	auto	Primary key
name	text	required	Location display name (e.g., “Hartford CT”)
salesTaxPercent	numeric	“0.00”	Sales tax rate for this location
createdAt	timestamp	now()	Creation timestamp

pushHistory Audit log for ServiceTitan pricebook push operations. Stores a snapshot of all settings used.

Column	Type	Default	Purpose
id	serial	auto	Primary key
tenantId	integer	required	Which tenant was pushed to
status	text (enum)	"pending"	pending / in_progress / completed / failed / cancelled / reverted
categoryIds	text (JSON)	required	JSON array of category IDs that were pushed
targetMePercent	numeric	required	M&E target used for this push
salesTaxPercent	numeric	required	Sales tax rate used
laborMode	text	required	"hours" or "percent"
laborHourlyRate	numeric	required	Hourly rate used
laborHours	numeric	"8"	Default hours per service
laborPercent	numeric	required	Labor % used
laborBurdenPercent	numeric	required	Labor burden % used
commissionPercent	numeric	required	Commission % used
commissionBurdenPercent	numeric	required	Commission burden % used
financingFeePercent	numeric	required	Financing fee % used
roundUp	boolean	false	Whether prices were rounded up
processedCount	integer	0	How many services were examined
updatedCount	integer	0	How many services were actually updated
skippedCount	integer	0	How many were skipped (no M&E)
errorCount	integer	0	How many failed
errorMessages	text (JSON)	null	JSON array of error message strings
originalValues	text (JSON)	null	JSON array of {serviceId, originalPrice, originalCost} for revert
createdAt	timestamp	now()	When push was initiated
startedAt	timestamp	null	When processing began
completedAt	timestamp	null	When processing finished

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3. System Calculator

File: `client/src/pages/Calculator.tsx` **Route**: `/`

The System Calculator computes a total selling price for an HVAC system installation based on materials, equipment, and a target M&E percentage.

Inputs

Input	Description
Target M&E %	Dropdown (10%–50% in 0.5% steps). What percentage of the final price should M&E represent.
Materials Cost	Dollar amount of materials
Equipment Cost	Dollar amount of equipment
Labor Hours or Labor %	Depends on labor mode in Settings
Subcontractor	Additional cost added to price
Crane	Additional cost added to price
Permit	Additional cost added to price
Discount	Dollar amount or % of total to subtract
OEM Rebate	Dollar amount added to revenue for GM% (not to price)
Utility Rebate	Dollar amount added to revenue for GM% (not to price)

Calculation Logic (in the `useEffect`, lines 89–175)

```
Step 1: Tax & M&E
  salesTaxRate = parseFloat(settings.salesTaxPercent) / 100
  tax = (materials + equipment) × salesTaxRate
  totalMe = materials + equipment + tax

Step 2: Base Price from M&E Target
  targetMeRate = parseFloat(targetMe) / 100
  basePrice = targetMeRate > 0 ? totalMe / targetMeRate : 0

Step 3: Add Other COGS, Apply Discount
  totalOtherCogs = subcontractor + crane + permit
  priceBeforeDiscount = basePrice + totalOtherCogs

  Discount Mode handling:
    If discountMode is "dollar":
      discountVal = discountInput (raw dollar amount)
    If discountMode is "percent":
      discountVal = priceBeforeDiscount × (discountInput / 100)

  price = max(0, priceBeforeDiscount - discountVal)

Step 4: Commission (always % of job, with burden)
  commissionRate = parseFloat(settings.commissionPercent) / 100
  commissionBurdenRate = parseFloat(settings.commissionBurdenPercent) / 100
  commissionCost = (price × commissionRate) × (1 + commissionBurdenRate)

Step 5: Financing
  financingRate = parseFloat(settings.financingFeePercent) / 100
  financingCost = price × financingRate

Step 6: Labor Cost (depends on settings.laborMode)
  laborBurdenRate = parseFloat(settings.laborBurdenPercent) / 100
  If laborMode is "hours":
    hourlyRate = parseFloat(settings.laborHourlyRate)
    laborCost = (hours × hourlyRate) × (1 + laborBurdenRate)
  If laborMode is "percent":
    laborPercentRate = parseFloat(laborPercent) / 100  (from user input, not settings)
    laborCost = (price × laborPercentRate) × (1 + laborBurdenRate)
```

```
Step 7: Estimated Labor Field (for ServiceTitan import)
  estimatedLabor = laborCost + commissionCost + tax + financingCost

Step 8: Display Price & Gross Margin
  displayPrice = roundUp ? Math.ceil(price) : price
  totalAllCogs = totalMe + commissionCost + financingCost + laborCost + totalOtherCogs
  effectiveRevenue = displayPrice + oemRebate + utilityRebate
  grossMarginAmount = effectiveRevenue - totalAllCogs
  grossMarginPercent = effectiveRevenue > 0 ? (grossMarginAmount / effectiveRevenue) × 100 : 0
  mePercent = displayPrice > 0 ? (totalMe / displayPrice) × 100 : 0
```

State Management

- All input values live in `CalculatorContext` so they persist across page navigations and can be shared with the Reverse Calculator.
- The `resetState` function resets to defaults while preserving targetMe and laborPercent from settings.

Additional Features

- **Round Up**: Toggle to round price to next whole dollar
- **Save Calc**: Opens dialog to name and save the calculation as a quote (stored as snapshot in `quotes` table)
- **Download Image**: Exports the results card as a PNG using `html-to-image`

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4. Repair Calculator

File: `client/src/pages/RepairCalculator.tsx` **Route**: `/repair`

The Repair Calculator prices individual repair jobs using material markup, labor, sales tax, commission, and financing.

Inputs

Input	Description
Parts Cost	Raw cost of parts
Material Markup	Percentage markup on parts+tax (dropdown: 10%–300%)
Labor Rate	Dollar per hour rate
Labor Hours	Number of hours
Burden %	Employer burden on labor/commission
Sales Tax %	Tax applied to parts
Commission %	As % of total selling price (backed in)
Financing Fee %	As % of total selling price
Highest Tech Pay Rate	For COGS calculation
Production Pay toggle	Switches between "$ + Hours" and "% of Job" labor modes
PSI Mode toggle	Preset values for PSI pricing

Calculation Logic (lines 67–134)

When labor mode is "$ + Hours" (default, `laborMode === "dollars"`):

```
salesTax = partsCost × taxRate
subtotal1 = partsCost + salesTax
```

```
markupAmount = subtotal1 × markupRate
subtotal2 = subtotal1 + markupAmount

directLaborCost = laborRate × laborHours
laborBurden = directLaborCost × burdenRate
burdenedLaborCost = directLaborCost + laborBurden
subtotal3 = subtotal2 + burdenedLaborCost

denominator = 1 - commissionRate × (1 + burdenRate) - financingRate
totalPriceCalc = denominator > 0 ? subtotal3 / denominator : subtotal3

directCommission = totalPriceCalc × commissionRate
commissionBurden = directCommission × burdenRate
burdenedCommission = directCommission + commissionBurden
financingFee = totalPriceCalc × financingRate

totalBeforeRounding = subtotal3 + burdenedCommission + financingFee
totalPrice = roundUp ? Math.ceil(totalBeforeRounding) : totalBeforeRounding
```

When labor mode is "% of Job" (`laborMode === "percent"`):

```
salesTax = partsCost × taxRate
subtotal1 = partsCost + salesTax
markupAmount = subtotal1 × markupRate
subtotal2 = subtotal1 + markupAmount

directLaborCost = 0 (commission handles everything)
subtotal3 = subtotal2

denominator = 1 - commissionRate × (1 + burdenRate) - financingRate
totalPriceCalc = denominator > 0 ? subtotal3 / denominator : subtotal3

directCommission = totalPriceCalc × commissionRate
commissionBurden = directCommission × burdenRate
burdenedCommission = directCommission + commissionBurden
financingFee = totalPriceCalc × financingRate

totalBeforeRounding = subtotal3 + burdenedCommission + financingFee
totalPrice = roundUp ? Math.ceil(totalBeforeRounding) : totalBeforeRounding
```

COGS & Gross Margin Calculation (lines 127–134)

```
techLaborCost = laborMode === "percent" ? 0 : highestTechRate × laborHours
burdenedTechLabor = techLaborCost × (1 + burdenRate)

COGS = partsCost + burdenedTechLabor + burdenedCommission
  (Note: Financing is excluded from COGS - it's a pass-through fee)

grossMargin = totalPrice - COGS
grossMarginPercent = totalPrice > 0 ? (grossMargin / totalPrice) × 100 : 0
```

Margin Health Thresholds (Repair Calculator, lines 138–146)

The Repair Calculator uses fixed thresholds (unlike System Calculator which uses `grossMarginMinPercent` from settings): - GM% >= 50 → "Strong" (green) - GM% >= 40 → "Caution" (amber) - GM% < 40 → "Low" (red)

State Sync

The Repair Calculator syncs its state to `CalculatorContext.repairState` via a `useEffect`, so the Reverse Calculator can copy values from it.

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5. Reverse Calculator

File: `client/src/pages/ReverseCalculator.tsx` **Route**: `/reverse`

The Reverse Calculator works backwards from a known selling price to determine the resulting gross margin percentage. It has two modes: **System** and **Repair**.

Mode Toggle

- Stored in `localStorage` key `pricebookcalc_reverse_mode`
- Persists per device/browser (not saved to database)
- Toggle UI: pill-style switcher at the top of the page

System Mode

Mirrors the System Calculator's inputs but replaces Target M&E % with a Selling Price input. Outputs GM% instead of computing a price.

Inputs: Selling Price, Materials, Equipment, Labor (hours or %), Subcontractor, Crane, Permit, Discount ($ or %), OEM Rebate, Utility Rebate

Calculation (lines 94–138):

```
tax = (materials + equipment) × salesTaxRate
totalMe = materials + equipment + tax
totalOtherCogs = subcontractor + crane + permit

Discount handling (differs from System Calculator):
  If discountMode is "percent": discountVal = sellingPrice × discountInput / 100
  If discountMode is "dollar": discountVal = discountInput
  price = max(0, sellingPrice - discountVal)

commissionCost = (price × commissionRate) × (1 + commissionBurdenRate)
financingCost = price × financingRate
laborCost = depends on laborMode (same as System Calculator)
estimatedLabor = laborCost + commissionCost + tax + financingCost

totalAllCogs = totalMe + commissionCost + financingCost + laborCost + totalOtherCogs
effectiveRevenue = price + oemRebate + utilityRebate
grossMarginAmount = effectiveRevenue - totalAllCogs
grossMarginPercent = effectiveRevenue > 0 ? (grossMarginAmount / effectiveRevenue) × 100 : 0
mePercent = effectiveRevenue > 0 ? (totalMe / effectiveRevenue) × 100 : 0
```

Repair Mode

Mirrors the Repair Calculator's inputs but adds a Selling Price input. Shows GM% based on entered price.

Inputs: Selling Price, Parts Cost, Material Markup, Labor Hours, Burden %, Sales Tax %, Commission %, Financing Fee %, Actual/Highest Tech Pay Rate, Production Pay toggle

Key difference from Repair Calculator: Uses user-entered selling price directly to compute commission/financing amounts, rather than deriving the price from costs. In Repair Calculator, the price is solved algebraically; in Reverse Repair mode, the price is given and used directly.

Calculation:

```
salesTax = partsCost × taxRate
subtotal1 = partsCost + salesTax            ← "Parts + Tax"
markupAmount = subtotal1 × markupRate
materialsCost = subtotal1 + markupAmount    ← "Parts + Tax + Markup"

If laborMode is "dollars":
  directLaborCost = highestTechRate × laborHours
  burdenedLaborCost = directLaborCost × (1 + burdenRate)
If laborMode is "percent":
  directLaborCost = 0
  burdenedLaborCost = 0

commission = sellingPrice × commissionRate × (1 + burdenRate)
financing  = sellingPrice × financingRate

COGS = subtotal1 + burdenedLaborCost        ← Parts + Tax + direct labor (no markup, no commission)
totalCosts = COGS + financing
grossMargin = sellingPrice - COGS
grossMarginPercent = sellingPrice > 0 ? (grossMargin / sellingPrice) × 100 : 0
mePercent = sellingPrice > 0 ? (subtotal1 / sellingPrice) × 100 : 0
```

Summary card values (repair mode): - **M&E %** = `subtotal1 / sellingPrice × 100` — percentage of (Parts + Tax) vs selling price. Markup is deliberately excluded so it reflects the true materials cost ratio.
- **Total Costs** = `COGS + financingFee` (not commission, not markup)

Price Comparison Chart (Repair Mode)

Replaces the Distribution pie chart. Shows two stacked vertical bars: **Calculated** (what the forward Repair Calculator formula would produce from the same inputs) and **Selling** (the user-entered price). Both bars share identical stacked segments so the visual difference is immediately clear.

Forward "Calculated Price" formula:

```
denominator = 1 - commissionRate × (1 + burdenRate) - financingRate
subtotal3 = materialsCost + burdenedLaborCost
calculatedPrice = subtotal3 / denominator
```

Stacked bar segments (same keys for both bars):

Segment	Color	Value — Calculated bar	Value — Selling bar
Parts + Tax	Gray	`subtotal1`	`subtotal1` (identical)
Markup	Blue	`markupAmount`	`markupAmount` (identical)
Labor	Green	`burdenedLaborCost`	`burdenedLaborCost` (identical)
Commission	Amber	`calculatedPrice × commissionRate × (1+burden)`	`sellingPrice × commissionRate × (1+burden)`
Financing	Violet	`calculatedPrice × financingRate`	`sellingPrice × financingRate`

Segment	Color	Value — Calculated bar	Value — Selling bar
Margin	Green/Red	`~0` (by formula construction)	`sellingPrice - all above`

The Margin segment is green when 0 and red when negative (selling below cost). A `LabelList` renders the total price above each bar.

No floating tooltip — instead a static 3-column comparison table (Segment | Calculated | Selling) sits below the chart, plus a delta callout (`+$X above / -$X below formula price`).

"Copy from" Buttons

- **System mode**: "Copy from System Calculator" — copies materials, equipment, labor, discount, rebates from the System Calculator's current state via `CalculatorContext.state`
- **Repair mode**: "Copy from Repair Calculator" — copies parts, markup, labor, fees from the Repair Calculator's current state via `CalculatorContext.repairState`
- Both buttons are always visible (not conditional on source data existing)

Reset

Each mode has its own reset function that clears all inputs to defaults.

---

6. Rebates (OEM & Utility)

Files: `client/src/pages/Calculator.tsx`, `client/src/pages/ReverseCalculator.tsx`, `client/src/components/Resul`
`client/src/contexts/CalculatorContext.tsx`

Rebates are dollar amounts that get added back to revenue for gross margin calculation without changing the selling price.

How It Works

- Two fields: OEM Rebate and Utility Rebate
- Present in both System Calculator and Reverse System mode
- Stored in `CalculatorContext.state` as `oemRebate` and `utilityRebate` strings

Impact on GM%

```
effectiveRevenue = displayPrice + oemRebate + utilityRebate
grossMarginAmount = effectiveRevenue - totalAllCogs
grossMarginPercent = (grossMarginAmount / effectiveRevenue) × 100
```

The selling price displayed to the customer does not change. Rebates only improve the GM% calculation because the company receives additional revenue from the manufacturer/utility.

Display

When rebate values are > 0, a "Rebates (Added to Revenue)" section appears in the ResultsCard cost breakdown with teal-colored highlight styling.

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7. ResultsCard Component

File: `client/src/components/ResultsCard.tsx`

Shared display component used by the System Calculator to show calculation results.

Props

Prop	Type	Description
`price`	number	Calculated selling price
`estimatedLabor`	number	Estimated labor field value
`breakdown`	object	Itemized costs (materials, equipment, tax, commission, etc.)
`metrics`	object	GM%, GM amount, M&E%
`salesTaxRate`	string	For display in breakdown
`discount`	number	Discount amount
`roundUp`	boolean	Whether rounding is enabled
`onRoundUpChange`	function	Toggle callback
`grossMarginMinPercent`	number	Threshold for margin health indicator
`oemRebate`	number	OEM rebate for display
`utilityRebate`	number	Utility rebate for display

Sections

1. **Primary Result Cards**: Price (with margin health bar) + Estimated Labor Field
2. **Metrics Row**: Gross Margin %, M&E %, Direct Labor $, Other COGS $
3. **Cost Breakdown**: Grouped into M&E, Direct Labor, Other COGS, and Rebates
4. **Pie Chart**: Visual distribution (Recharts) with labeled slices

Margin Health Indicator

```
if GM% >= grossMarginMinPercent → "Strong" (green)
if GM% >= grossMarginMinPercent - 2 → "Caution" (amber)
otherwise → "Low" (red)
```

---

8. Calculator Context

File: `client/src/contexts/CalculatorContext.tsx`

React Context that provides shared state between all three calculators.

State Shape

```tsx
interface CalculatorState {
  materials: string;
  equipment: string;
  laborHours: string;
  laborPercent: string;
  targetMe: string;
  discount: string;
  discountMode: "dollar" | "percent";
  subcontractor: string;
  crane: string;
```

```
  permit: string;
  roundUp: boolean;
  oemRebate: string;
  utilityRebate: string;
}

interface RepairCalculatorState {
  partsCost: string;
  materialMarkup: string;
  laborRate: string;
  laborHours: string;
  burdenPercent: string;
  salesTaxPercent: string;
  commissionPercent: string;
  financingFeePercent: string;
  highestTechRate: string;
  laborMode: "dollars" | "percent";
}
```

How It's Used

- **System Calculator** reads/writes `state`
- **Repair Calculator** writes to `repairState` via `useEffect` (syncs local state to context)
- **Reverse Calculator** reads `state` and `repairState` for "Copy from" functionality
- Wrapped at the app root in `App.tsx` via `<CalculatorProvider>`

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9. Settings & Per-Device Cache

Files: `client/src/pages/Settings.tsx`, `client/src/hooks/use-settings.ts`, `server/routes.ts`, `server/storage.ts`

Settings Page (`/settings`)

Provides a form to configure all calculator defaults: Target M&E %, Sales Tax, Labor mode/rate, Commission, Financing, Burden rates, and Minimum GM%.

Per-Device Cache

How it works: 1. When user saves settings, values are written to both the database AND `localStorage` (key: `pricebookcalc_settings`) 2. On page load, the `use-settings` hook checks `localStorage` first 3. If local settings exist, they override database values 4. If no local settings exist, database settings are used as defaults 5. This means each device/browser can have its own independent settings

API Endpoints

- `GET /api/settings` — Fetch current settings
- `PUT /api/settings` — Update settings

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10. Quote Saving & History

Files: `client/src/pages/Calculator.tsx`, `client/src/pages/AutoMECalculator.tsx`, `client/src/pages/RepairCalcu` `client/src/pages/RepairGMCalculator.tsx` (save dialogs / Reveal Price gate), `client/src/pages/History.tsx`, `client/src/hooks/use-tenants.ts`, `server/routes.ts`, `server/storage.ts`

Saving a Quote

From any of the four calculators (System GM, System M&E, Repair, Repair GM), clicking "Save Calc" (or confirming the Reveal Price dialog — see below) saves the current calculated results as a snapshot: - Input values (materials, equipment, labor hours, subcontractor, crane, permit) - Output values (price, GM%, estimated labor, M&E%) - `tenantId` — the tenant currently selected in that calculator's `TenantTaxSelector`, if any (nullable; used later for filtering on the History page)

Reveal Price Gate

For roles `everyone` and `user` (not `manager`/`admin`), the calculated results on System GM, System M&E, Repair, and Repair GM are hidden behind a "Reveal Price" button (`RevealPriceGate` component). Confirming the dialog (Job Number required, Customer Name optional) saves a quote — title = job number, `customerName` set — without generating a PNG, then reveals the results. The Reverse Calculator is excluded (price is entered directly, nothing to reveal).

History Page (`/history`)

Displays all saved quotes in a card list with: - Quote title, customer name (if set) - Calculated price, GM% - Date saved, added-by user - Tenant name badge (only shown when the quote has a resolvable `tenantId`) - Sort control (newest/oldest/price, etc.) - **Tenant filter dropdown** — only rendered when the current user has at least one tenant available. Options are "All Tenants" plus every tenant returned by `useTenants()` (which calls `GET /api/tenants`, already role-scoped server-side: admins see every tenant, `user`/`manager` see only tenants granted via `user_tenant_access`). The filter is a pure client-side filter over the already-authorized quote list — no additional backend query parameter is needed since the available tenant options are already scoped correctly. Quotes with `tenantId = null` (legacy, saved before the column existed) only appear under "All Tenants". - Delete action

API Endpoints

- `GET /api/quotes` — List quotes (admin sees all, user sees own — scoped by `userId`, independent of the tenant filter which is applied client-side)
- `POST /api/quotes` — Create new quote (accepts optional `tenantId`)
- `DELETE /api/quotes/:id` — Delete a quote
- `GET /api/tenants` — Role-scoped list of tenants; reused directly to populate the History page's tenant filter options

Admin: Clear History

In the Admin section of the navigation (after unlocking with PIN), there's a "Clear All History" option that deletes all saved quotes.

---

11. ServiceTitan Integration

Files: `server/servicetitan.ts`, `server/routes.ts`

ServiceTitanAPI Class

Located in `server/servicetitan.ts`, this class handles all communication with the ServiceTitan Pricebook v2 API.

Authentication:

```
OAuth2 client_credentials flow
POST https://auth.servicetitan.io/connect/token
Tokens cached in-memory for 15 minutes per tenant
```

Key Methods: | Method | Purpose | |——–|———| | `getAccessToken()` | Obtain/refresh OAuth token | | `getCategories()` | Fetch pricebook category tree | | `getServices(categoryId)` | Fetch services by category | | `getMaterials(onProgress)` | Fetch all materials (paginated) | | `getEquipment(onProgress)` | Fetch all equipment (paginated) | | `updateService(id, data)` | Update service price/hours | | `createService(data)` | Create new service | | `uploadServiceImage(id, file)` | Upload image to service | | `getGLAccounts()` | Fetch GL account list |

API Field Mapping (ServiceTitan PricebookCalc)

ServiceTitan Field	API Property	Updatable?
Static Price	`price`	Yes
Estimated Labor Cost	`cost`	No (calculated by ST from Hours × internal rate)
Hours	`hours`	Yes

Tenant Management

Page: `client/src/pages/Tenants.tsx` (Route: `/tenants`, protected) - CRUD operations for ServiceTitan tenant credentials - Credentials are masked in API responses (never sent to frontend in full) - Each tenant has: name, tenantId, clientId, clientSecret, appKey - Tenants can be assigned a location (for sales tax) - Sort order controls display ordering

API Endpoints

- `GET /api/tenants` — List all tenants (credentials masked)
- `POST /api/tenants` — Create tenant
- `PUT /api/tenants/:id` — Update tenant
- `DELETE /api/tenants/:id` — Delete tenant

---

12. Push to ServiceTitan (Update Pricebook)

Files: `client/src/pages/PushToServiceTitan.tsx`, `server/pushProcessor.ts`, `server/routes.ts` **Route**: `/push` (password-protected)

The "Update Pricebook" tab on the Push to ServiceTitan page lets users push calculated M&E pricing to existing services in ServiceTitan. The page also includes a collapsible **Tax Regions** management section (between tenant selection and the tabs) for inline add/edit/delete of tax regions without navigating away.

How It Works

1. **User selects**: Tenant, categories, and confirms settings
2. **Backend receives push request** and creates a `pushHistory` record with status "pending"
3. **Backend responds immediately** (fire-and-forget) — processing happens in background
4. **Push processor** (`server/pushProcessor.ts`) runs:
   - Fetches services from selected categories
   - Fetches all materials and equipment for cost lookup
   - For each service with M&E > $0:
     - Calculates `newPrice = totalMe / (targetMePercent / 100)`
     - Calculates `estimatedLabor = laborCost + commissionCost + tax + financingCost`
     - Updates service `price` via API (cost is read-only in ServiceTitan)
   - Stores original values for potential revert

Price Calculation per Service

```
materialsCost = sum of (material.cost × material.quantity) for linked materials
equipmentCost = sum of (equipment.cost × equipment.quantity) for linked equipment
rawMeCost = materialsCost + equipmentCost
tax = rawMeCost × salesTaxRate
totalMe = rawMeCost + tax
newPrice = totalMe / (targetMePercent / 100)
```

Category Filtering

The push processor intelligently handles parent-child category relationships: - Fetches the full category tree to understand hierarchy - If a parent category AND its child are both selected, only the child is queried (parent query would include children redundantly) - Services are filtered to only include those whose category exactly matches a selected category

Progress Tracking

Uses Server-Sent Events (SSE) via `server/pushProgress.ts`: - Client opens `EventSource` connection to `/api/servicetitan/:tenantId/push/progress` - Server emits progress events for phases: `services` → `materials` → `equipment` → `updating` → `complete` - **Heartbeat**: Server writes a `: ping` comment every 25 seconds to prevent proxy idle-timeout disconnects - **Current state on connect**: On every (re)connect the server immediately sends the latest known progress for that tenant, so a reconnecting client is instantly up to date - **Reconnect**: The frontend's `onerror` handler does NOT close the EventSource — the browser's built-in SSE auto-reconnect (3-second delay, set via `retry:` field) handles drops transparently

Revert

Push history records store `originalValues` (JSON array of original prices). The revert endpoint (`POST /api/push-history/:id/revert`) restores these values via the same `updateService` API call.

Retry

Any push history record in a terminal state (`completed`, `failed`, `cancelled`, or `reverted`) can be retried.

Endpoint: `POST /api/push-history/:id/retry`

- Reads **all** stored settings from the original history record (categoryIds, targetMePercent, salesTaxPercent, laborMode, labor rates, commission, burden, financing, roundUp)
- Creates a **new** history record (original is preserved unchanged)
- Follows the same queue/concurrency path as a fresh push — returns `{ historyId, queued }`
- Frontend shows a blue **Retry** ( ) button on every finished record; spinner while pending

Push History Error Display

Error messages in the history list are shown in full (`break-all` for long lines). When a record has more than 3 errors, a **"Show X more / Show less"** toggle expands or collapses the full list.

---

13. Push Queue & Rate Limiting

File: `server/pushQueue.ts`

Concurrency Control

- Maximum 3 concurrent pushes (`MAX_CONCURRENT_PUSHES = 3`)
- Each push processes up to 20 service updates in parallel (`CONCURRENCY = 20` in pushProcessor)

- Total maximum: 20 × 3 = 60 requests/second (within ServiceTitan's 60 req/sec limit)

Queue Behavior

- If 3 pushes are already active, new pushes are queued
- Queue processes in FIFO order
- When an active push completes, the next queued push starts automatically
- `pushQueue` extends `EventEmitter` — emits 'ready' events to trigger dequeued pushes

Exponential Backoff Retry

`withRetry` in `server/pushQueue.ts` retries transient ServiceTitan errors automatically (max 3 attempts):

Error category	Condition	Base delay
Rate limit	HTTP 429 / "rate limit" / "too many requests"	1 s
Gateway error	HTTP 500 / 502 / 503 / 504 (incl. Cloudflare 500)	2 s
Request timeout	"timeout after 30s" / AbortError / ECONNRESET	5 s
Snapshot isolation	"snapshot isolation" / "ddl statement"	2 s

Each attempt waits `baseDelay × 2^attempt` before the next try. Any error not matching these categories is thrown immediately without retry.

Queue Status API

- `GET /api/servicetitan/push-queue/status` — Returns active and queued push info

---

14. New System Task & New Repair Task

File: `client/src/pages/PushToServiceTitan.tsx` (tabs 2 and 3)

New System Task Tab

Creates new system installation services in ServiceTitan with M&E-based pricing.

Includes: - Price calculator (Materials/Equipment cost → price via M&E target) - Form fields: Code, Name, Description, Warranty Description - Checkboxes: Taxable, Allow Discount Codes, Allow Membership Discounts - Hours field, GL Account dropdown (searchable), Category selection - Image upload support - PSI Mode toggle

New Repair Task Tab

Creates new repair services in ServiceTitan with labor/commission pricing.

Includes: - Repair price calculator (Parts, Markup, Labor, Tax, Commission, Financing) - Same form fields as System Task - Production Pay toggle ($ + Hours vs % of Job) - PSI Mode toggle - Real-time price breakdown with gross margin indicator

API Endpoints

- `GET /api/servicetitan/:tenantId/gl-accounts` — Fetch GL accounts for dropdown
- `POST /api/servicetitan/:tenantId/services` — Create new service
- `POST /api/servicetitan/:tenantId/services/:serviceId/image` — Upload image to service

---

15. PSI Mode Toggle

File: `client/src/pages/PushToServiceTitan.tsx`, `client/src/pages/RepairCalculator.tsx`

PSI Mode is a quick-preset toggle that applies a specific set of values for PSI (Performance Service Institute) pricing standards.

New System Task PSI Mode

- Switches GL Account from "Revenue" (default) to "Sales"

New Repair Task / Repair Calculator PSI Mode

When PSI ON: | Setting | Value | |———|——-| | Material Markup | 40% | | Labor Rate | $600 | | Sales Tax | 7.4% | | GL Account | "Sales" |

When PSI OFF (defaults): | Setting | Value | |———|——-| | Material Markup | 200% | | Labor Rate | $175 (from settings) | | Sales Tax | From tenant's assigned location (fallback 6.35%) | | GL Account | "Revenue" | | Financing Fee | 4% |

---

16. Location-Based Sales Tax (Tax Regions)

Files: `client/src/pages/Locations.tsx`, `client/src/pages/PushToServiceTitan.tsx`, `server/routes.ts`, `server/storage.ts`, `shared/schema.ts` **Route**: `/locations` (password-protected admin page), also accessible inline on `/push`

Purpose

A reference library that maps location names (tax regions) to their local sales tax percentages. Tenants can be assigned a location, and when that tenant is selected on the Push to ServiceTitan page, the tax rate auto-populates into **all three tabs** (Update Pricebook, New System Task, New Repair Task).

Managing Tax Regions

Tax regions can be managed in two places: 1. **Admin Locations page** (`/locations`) — full-page CRUD management 2. **Push to ServiceTitan page** (`/push`) — collapsible inline "Tax Regions" section between tenant selection and the main tabs, allowing add/edit/delete without navigating away

Both interfaces share the same `/api/locations` endpoints and data.

Seeded Locations

Location	Sales Tax
Gaithersburg MD	6.00%
Hartford CT	6.35%
Gypsum CO	7.40%
Jupiter FL	7.00%
Denver CO	9.15%

Location	Sales Tax
Houston TX	8.25%

How Tax Auto-Populates

1. User selects a tenant on the Push to ServiceTitan page
2. If that tenant has an assigned `locationId`, the location's `salesTaxPercent` is fetched
3. The tax rate is set on **both** `salesTaxPercent` (used by Update and New System tabs) and `repairSalesTaxPercent` (used by New Repair tab)
4. If PSI mode is toggled off, the location-based tax is restored
5. Fallback: global settings value, or 6.35% if no location is assigned

API Endpoints

- `GET /api/locations` — List all locations
- `POST /api/locations` — Create location
- `PUT /api/locations/:id` — Update location
- `DELETE /api/locations/:id` — Delete location (also clears `locationId` on any associated tenants)

---

17. Commission & Labor Modes

Commission

Commission is **always** calculated as "% of Job" — it's a percentage of the total selling price that gets backed into the total. There is no toggle for commission mode; it's always this way.

```
directCommission = totalPrice × commissionRate
commissionBurden = directCommission × burdenRate
burdenedCommission = directCommission + commissionBurden
```

Labor Mode (Production Pay Toggle)

Files: `client/src/pages/RepairCalculator.tsx`, `client/src/pages/ReverseCalculator.tsx`, `client/src/pages/PushToServiceTitan.tsx`

A Switch component labeled "Production Pay" toggles between two modes:

"$ + Hours" Mode (default)

- Labor Cost = Labor Rate × Labor Hours
- Commission is a separate % of job
- Both labor and commission have burden applied independently
- Formula: `T = subtotal3 / (1 - commissionRate×(1+burdenRate) - financingRate)`

"% of Job" Mode

- Labor cost is $0 — commission handles everything (labor + commission combined)
- No separate labor inputs shown
- Commission % field covers both labor and commission
- Formula: `T = subtotal2 / (1 - commissionRate×(1+burdenRate) - financingRate)`
- Price breakdown hides Labor line and relabels Commission as "Commission incl. Labor"
- Highest Tech Pay Rate and COGS tech labor are zeroed (no hours to calculate)

---

18. Authentication & Access Control

ServiceTitan Section (Password)

Files: `client/src/contexts/ServiceTitanAuthContext.tsx`, `client/src/components/PasswordGate.tsx`, `server/routes.ts`

Protected pages: Push to ServiceTitan, Push History, Tenants, Locations

How it works: 1. Pages are wrapped in `<PasswordGate>` component in `App.tsx` 2. `PasswordGate` checks `useServiceTitanAuth()` context 3. If not unlocked, shows a password prompt card 4. Password is verified server-side against `settings.serviceTitanPassword` 5. On success, the client's IP is stored in a server-side `Map` with 8-hour expiration 6. Subsequent requests from the same IP are auto-authenticated

Server-side IP tracking:

```
const unlockedIPs = new Map<string, number>(); // IP -> expiration timestamp
const AUTH_DURATION = 8 * 60 * 60 * 1000; // 8 hours
```

API Endpoints: - `GET /api/admin/check-auth` — Check if current IP is authenticated - `POST /api/admin/verify-password` — Verify password and unlock IP - `POST /api/admin/change-password` — Change the ServiceTitan password

Admin Section (PIN)

Files: `client/src/contexts/AdminAuthContext.tsx`, `client/src/components/Navigation.tsx`

Protected features: Change Password, Change PIN, Clear History, Lock Push Section

How it works: 1. Admin section is in the sidebar navigation 2. Expanding it shows a PIN input form 3. PIN is verified server-side against `settings.adminPin` 4. Same IP-based authentication with 8-hour expiration 5. Once unlocked, admin actions are available inline in the sidebar

Admin capabilities: - Change ServiceTitan password - Change Admin PIN - Clear all saved quote history - Lock/unlock the ServiceTitan push section

API Endpoints: - `GET /api/admin/check-admin-auth` — Check admin auth status - `POST /api/admin/verify-pin` — Verify PIN - `POST /api/admin/change-pin` — Change admin PIN - `DELETE /api/admin/clear-history` — Delete all quotes

---

19. Version Check Banner

File: `client/src/components/VersionCheck.tsx`, `server/routes.ts`

Notifies users when a new version of the app has been published.

How It Works

1. Server generates a unique build ID (`Date.now().toString()`) at startup
2. `/api/version` endpoint returns this build ID
3. Frontend component polls `/api/version` every 30 seconds
4. On first load, it stores the build ID in a `useRef`
5. If a subsequent poll returns a different build ID, a sticky banner appears
6. Banner says "A new version of PricebookCalc is available" with a "Refresh Now" button
7. Button calls `window.location.reload()` to load the new version

Styling

- Sticky banner at the very top of the page (`z-index: 9999`)

- Primary background color with spinning refresh icon
- Appears on all pages (component is rendered at the app root level)

---

20. Navigation & Routing

File: `client/src/components/Navigation.tsx`, `client/src/App.tsx`

Route Map

Path	Page	Protection
`/`	System Calculator	None
`/repair`	Repair Calculator	None
`/reverse`	Reverse Calculator	None
`/push`	Push to ServiceTitan	Password
`/push-history`	Push History	Password
`/tenants`	Manage Tenants	Password
`/locations`	Manage Locations	Password
`/history`	Saved Quotes	None
`/settings`	Calculator Settings	None

Navigation Component

- **Desktop**: Fixed sidebar on the left (264px wide)
- **Mobile**: Hamburger menu that slides in from the left
- **ServiceTitan section**: Collapsible group with lock icon when locked
- **Admin section**: Collapsible group at bottom with PIN-gated access

Provider Hierarchy (App.tsx)

```
QueryClientProvider
    TooltipProvider
        ServiceTitanAuthProvider
            AdminAuthProvider
                CalculatorProvider
                    VersionCheck
                    Toaster
                    Router
                        Navigation (sidebar)
                        <main> (page content)
```

Appendix: Common Patterns

InputMoney Component

File: `client/src/components/InputMoney.tsx`

Reusable input with dollar sign prefix, consistent styling, and label. Used across all calculators for currency/numeric inputs.

- Accepts `type` prop: `"currency"` ($ prefix), `"percent"` (% prefix), or `"number"` (# prefix)

- Defaults `placeholder` to `"0"` when no explicit placeholder is passed — all numeric fields across every page show a light-gray "0" hint when empty rather than appearing blank or hard-coding an actual zero value
- Spreads all standard HTML input attributes; individual call sites can override `placeholder` with their own value (e.g., `placeholder="6.35"` for a pre-seeded tax rate hint)

TanStack Query Usage

- Queries use default `queryFn` (no need to define fetch logic)
- Mutations use `apiRequest` from `@/lib/queryClient`
- Cache invalidation by `queryKey` after mutations
- Hierarchical keys: `['/api/resource', id]` for proper invalidation

Drizzle ORM Pattern

- Schema defined in `shared/schema.ts`
- Insert schemas created with `createInsertSchema` from `drizzle-zod`
- Storage operations in `server/storage.ts` implement `IStorage` interface
- All CRUD routed through storage interface (routes stay thin)